From Screen Reading to Aural Glancing: Towards Instant Access to Key Page Sections

Prathik Gadde and Davide Bolchini
Indiana University School of Informatics and Computing - Indianapolis
535 West Michigan Street, Indianapolis, Indiana 46202
[pgadde, dbolchin]@iupui.edu

ABSTRACT

Whereas glancing at a web page is crucial for navigation, screen readers force users to listen to content serially. This hampers efficient browsing of complex pages and maintains an accessibility divide between sighted and screen-reader users. To address this problem, we adopt a three-pronged strategy: (1) in a user study, we identified key page-level navigation problems that screen-reader users face while browsing a complex site; (2) through a crowd-sourcing system, we prioritized the most relevant sections of different page types necessary to support basic tasks; (3) we introduced DASX, a navigation approach that augments the ability of screen-reader users to "aurally glance" at a complex page by accessing at any time the most relevant page sections. In a preliminary evaluation, DASX markedly reduced the gap in page navigation efficiency between screen-reader and sighted users. Our contribution provides the groundwork for rethinking access strategies that strongly tie aural navigation to user's tasks.

Categories and Subject Descriptors

H.5.2 [**Information Interfaces and Presentation**]: User Interfaces – *Interaction styles;* K.4.2 [Computers and Society]: Social Issues – *Assistive technologies for person with disabilities.*

Keywords

Web navigation; voice browsing; fast browsing; direct web access; eCommerce web applications; blind users; screen-reader users.

1. INTRODUCTION

> *On Amazon.com, a screen-reader user wants to buy a new laptop with good customer reviews. After sifting through the search results, the user clicks on a laptop to know more. After double-checking the product details, the user looks for customer reviews. To do so, the user is forced to listen carefully to at least four page sections following the product details (e.g., "customers who bought this item also bought," "frequently bought together"). If the reviews section is missed, all previous sections need to be re-listened to. Finally, upon deciding to buy the laptop, the user goes back to the very top of the page to add the item to the cart.*

This scenario highlights two main problems surrounding screen-reader navigation in large, complex websites. First, *it is difficult to quickly identify the available sections of a page.* Whereas a page visually communicates extensive information at once, including content, overall semantics, orientation cues, and navigation possibilities, the linear reading strategies of screen readers make such multidimensional communication difficult or even impossible. What could be the aural equivalent to the fundamental act of "glancing at a page"? Second, even when users know the relevant page sections, there is currently no efficient navigation mechanism *to access that section quickly* besides trying to search for a keyword that matches the content. Screen-reader users rely on a few workarounds to break this rigidly serial reading mode. "Skip to main content," "list of links" on the page, or "searching" for a word, are examples of attempts to support direct access. Yet, because screen readers are agnostic with respect to the different structures of web pages, these strategies often fall short. For example, whereas "skip to main content" avoids the need to listen to repeating navigation headings, it can also cause serious navigation problems, such as skipping *sort* and *filter options* during search.

In this paper, we propose to augment screen reading with a novel navigation approach: instead of being forced to listen to page items serially, screen-reader users can quickly browse a selection of highly-relevant page sections through always available, dialogically-inspired commands, activated via keystrokes or voice. For example, let us imagine a user in the "product detail" page asking *What's there?* and instantly being able to access the five most important sections of that page (out of the over 30 available), including the customer reviews, which can then be selected with minimal effort. Our contribution is threefold:

- A formative user study (Section 3) with eight blind and visually impaired (BVI) and four sighted users investigated and compared web navigation experience and strategies in a complex site. We uncovered basic problems, at the page level, that account for the large amount of time and mechanical effort necessary for screen-reader users to complete even simple tasks.
- Through a custom-developed crowd-ranking system, we identified and prioritized the most important page sections for key shopping tasks (Section 4). We showed that the page sections considered relevant to the task at hand are often very difficult for screen-reader users to access.
- We introduce DASX (Section 5), a navigation approach that empowers screen-reader users to access directly the most important sections of a page. We exemplify our approach for shopping sites; in an analytical evaluation of an expert user's task performance, DASX not only accelerated screen-reader navigation but also greatly reduced keyboard usage and link clicking.

By contributing a deeper understanding of strategies for accessible, aural glancing, our work paves the way for augmenting the ability of screen-reader users to quickly get a sense of what's available on the page and accelerate access to relevant portions. Our approach can be also extended to mobile accessibility.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ASSETS'14, October 20–22, 2014, Rochester, NY, USA.
Copyright © 2014 ACM 978-1-4503-2720-6/14/10…$15.00.
http://dx.doi.org/10.1145/2661334.2661363

2. RELATED WORK

Because of its broad adoption, screen-reader navigation receives much attention from the accessibility research community. Specifically, our work joins forces with efforts aimed at *overcoming the serial nature* of screen-reader browsing during the *page-level* navigation of large, information-intensive websites.

The Web Content Accessibility Guidelines has recognized the importance of quickly targeting key page sections in large websites during screen-reader navigation [22]. Whereas Guidelines 1.3 and 2.4 emphasize the importance of presenting multiple access structures to the content, Guideline 2.4.1 specifically dictates providing mechanisms to bypass blocks of repeating content over multiple pages. Such seminal effort spurred the adoption of the "skip to content" link on many sites. This is an important first step towards more efficient page navigation for screen-reader users, but it assumes that a generic content section (however is defined) is always the primary target of interest for the user. Yet users might need direct access to different sections based on the nature of the page at hand (e.g., list page, home, product page, or order page).

Fundamental work focused on developing techniques to reduce the complexity of web pages to enhance accessibility. For example, important and highly-novel accessibility solutions have been developed by partitioning pages [7][15], extracting content based on the visual design [6], directly navigating from a page to the relevant content in the target page [3], annotate webpages with structural and navigational ontologies to enhance mobility support [21], refactoring content to make it more suitable for aural consumption [12], and thematically organizing content for fast screen-reader navigation [13]. While most of these contributions operate on the content structure of the site to make it more accessible, they can be complemented by new ways of accelerating access to the pages as they are, thus rethinking aural access without adapting the application itself.

To bridge the divide between screen-reader users and sighted users, other important research provided users with the ability to get notified about page updates that could be otherwise easily missed [4], automate repetitive tasks like bill payments [14], and provide an additional layer of "whispered" suggestions while browsing [23]. Recent work has also contributed techniques to support non-visual skimming of the content [1]. With this technique, screen-reader users can access automatic summaries of online articles to read content much faster. We argue that accessible skimming can be complemented by aural glancing, i.e. the ability to make sense of the most important sections of a complex page (which includes different types of content) and quickly access them on-demand.

To support faster screen-reader navigation, prior work has also focused on identifying and validating aural strategies to accelerate *inter-page navigation* of large collections, including bypassing lists with guided tours [10][20] and back navigation shortcuts [18]. This body of work can be extended with solutions that tackle the basic problem of having to serially navigate a large amount of *page-level sections* to make sense of the content available and immediately target relevant segments on-demand.

Generations of advanced aural browsers and strategies have also been explored in the past two decades. Examples include PWWebSpeak [9], IBM Home Page Reader [2], Emacs W3 [17], Pipebeach [5], Conversa [11], and TeleBrowse [16]. Unlike screen readers, which read out the web content serially, these browsers simplified access strategies by providing users with the ability to customize a web page based on different features. With the growth of the web, these solutions face new problems stemming from the growing complexity of today's highly-interactive and information-overloaded web applications.

By recognizing the potential of voice access to support more dialogic, direct browsing of page content, the Voice Browser Working Group has also developed and standardized several technologies [19]. Further, browsers like Opera offer voice-controlled navigation that supports basic operations, such as *opening new tabs, closing tabs, reloading documents, and going to the previous page* [24]. While these functions are key to inter-application navigation, they do not provide techniques for efficient page-level navigation in complex sites.

3. FIELD STUDY

3.1 Purpose and Participants

The goal of our formative user study was twofold: (1) to uncover and characterize page-level navigation behavior, issues, and strategies of screen-reader users while browsing a large website like Amazon.com for basic shopping tasks; and (2) to gauge, both qualitatively and quantitatively, the navigation performance gap between screen-readers and sighted users.

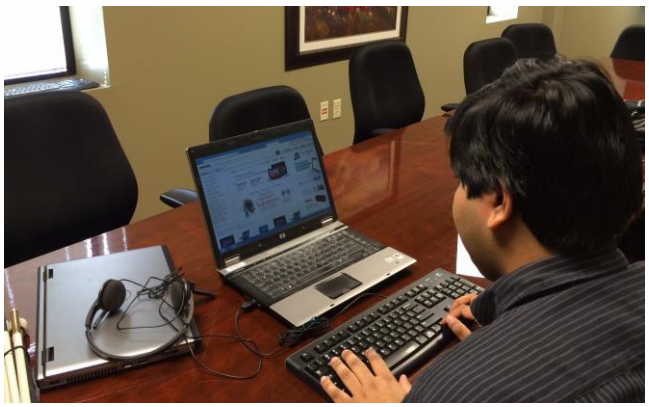


Figure 1. Screen-reader users performed basic tasks on Amazon.com and discussed their page-level navigation experience.

We conducted semi-structured interviews with screen-reader users and performed contextual observations of navigation behavior (Figure 1) with a focus on the following questions:

- What are the major obstacles for efficient intra-page navigation in large, information-intensive sites?
- To which extent does the linear navigation mechanics of screen readers slow down access for BVI users?
- How does the structure of different page types impact navigation needs and behavior?

For the purpose of this exploratory study, we selected Amazon.com because it exhibits a level of page density and complexity that is increasingly adopted by most eCommerce sites. We recruited eight BVI (aged 24-62; 5 male, 3 female) and four sighted (aged 22-26; 4 male) participants from a large Midwestern US city. All BVI participants were expert screen-reader users with at least four years of experience.

3.2 Procedure and Setting

After a brief introductory session about the study, all participants were invited to perform a training task, which enabled them to refresh their memory of using Amazon.com. Then they performed two test tasks. All sessions were organized as semi-structured interviews with the observer asking questions on critical incidents

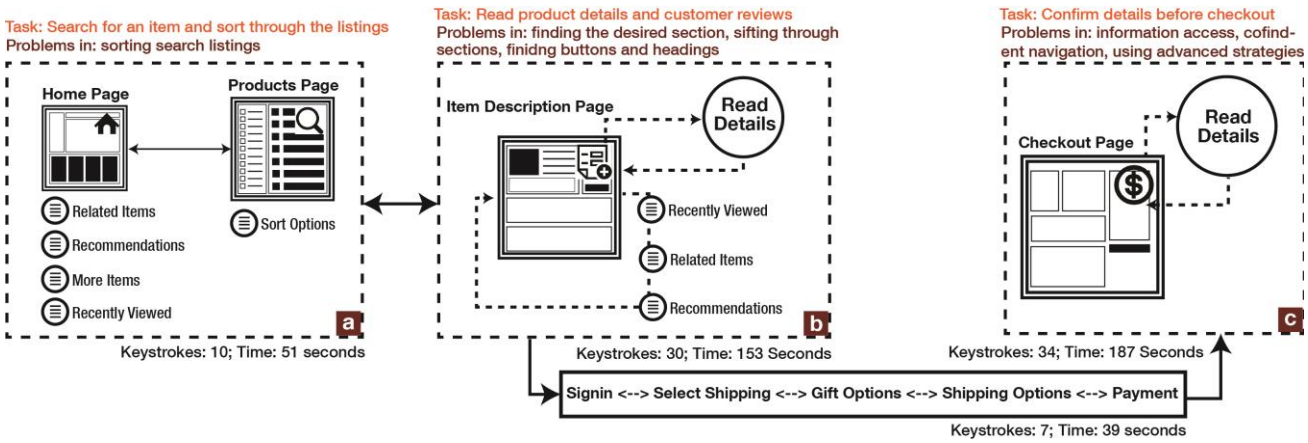


Figure 2. Users encountered problems in three key navigation contexts: (a) Moving from the home to the search results and efficiently sifting through them; (b) Browsing product details and reviews while bombarded by additional page sections; (c) Trying to quickly and frequently double-checking order details before making a confident purchase decision.

and stopping points during navigation. Participants were instructed to think aloud while performing the following two tasks, which sampled the depth and breadth of the website:

> ***Task 1: Search and Compare.*** *You would like to buy a Dell laptop with at least a 500 GB hard drive, i5 processor, and 6 GB RAM. Your budget is $800. Find and compare at least three models. Select the laptop that best meets the above-mentioned criteria. Be prepared to justify the selected model.*
>
> ***Task 2: Select and Checkout Review.*** *Your friend's birthday is coming up and you would like to give her an Amazon gift card. Without using the "search," purchase a $50 gift card. Select mail as the delivery method and add it to the cart. Proceed to checkout until you reach the screen with a "place your order" button. Double-check your order, shipping, and payment details.*

To make the participants comfortable while performing the tasks, we let them choose the study location. Ten of 12 sessions were conducted either at the participant's workplace or home. The remaining two sessions were conducted in our lab. All participants performed the training and the tasks on a laptop (minimum PC requirements: 1 GB RAM, Core 2 Duo Processor, 650 MB HDD, MS Windows XP or later) with the JAWS 15.0 screen reader and a browser of their choice. For approximately two hours of participation, each participant received a $20 gift card.

3.3 Data Collection and Analysis

We used *uLoglite* [30] to record the participants' keystrokes and time spent on each page, and we audio-recorded all sessions for transcription and analysis. The recorded audio files of the sessions were manually examined and annotated to identify explanations for the navigation problems encountered and salient instances of user's behavior. *uLoglite* recordings were examined to collect the following performance measures: user actions (including keystrokes for screen-reader users and scrolling and clicks for sighted users); time spent on each page and overall time on task. Because we adopted a thinking-aloud protocol, the time spent answering questions was removed from the time on task during data analysis. To gain further insights into the qualitative feedback, we also abstracted the low-level mechanics of the page browsing to a higher-level model that highlights the navigation contexts associated to most issues emerged (Figure 2).

4. Findings

Overall, we identified major page-level navigation issues for screen-reader users that revolve around seven main themes:

(1) Even simple pages, by structure, pose navigation problems;
(2) Rigid access to different page types hampers navigation;
(3) Fear of missing crucial information forces unnecessary mechanical and cognitive effort;
(4) Little tolerance of navigation mistakes demands careful attention and line-to-line reading;
(5) Difficulty to distinguish sections and overloaded pages obstruct access to the key section;
(6) Shortcuts like 'Skip to Content' are not always optimal;
(7) Difficulty in re-finding information forces users to remember details and unnecessarily overloads short term memory;

The rest of this section elaborates on each theme.

4.1 Even Simple Pages Obstruct Navigation

Surprisingly, all screen-reader users had more troubles navigating through the checkout page than all other pages, even though the checkout page has far fewer sections and exhibits a much simpler layout. When asked about the reason for this, participants pointed out that it was very difficult to understand how the page sections were organized. This hindered their ability to quickly identify crucial information to review their order (making sure they have all and only the right products in the cart). One participant said:

> *[P1] "I don't want to order without knowing what is there in my cart. This page is completely different from the other pages and I couldn't find what is there in my cart. After I finished everything, to sit in the checkout page and not finding the information I am ordering is frustrating."*

By examining the user performance measures, we noticed that it took more steps and time for screen-reader users to navigate through a simpler page, by structure, than a relatively complex page. While browsing through a complex *item description* page, using advanced features, screen-reader users were able to quickly achieve their information needs (Actions: 28, Time on task: 142 s). However, when navigating a simpler *checkout* page they found it extremely difficult to find information they wanted (Actions: 34, Time on task: 187 seconds).

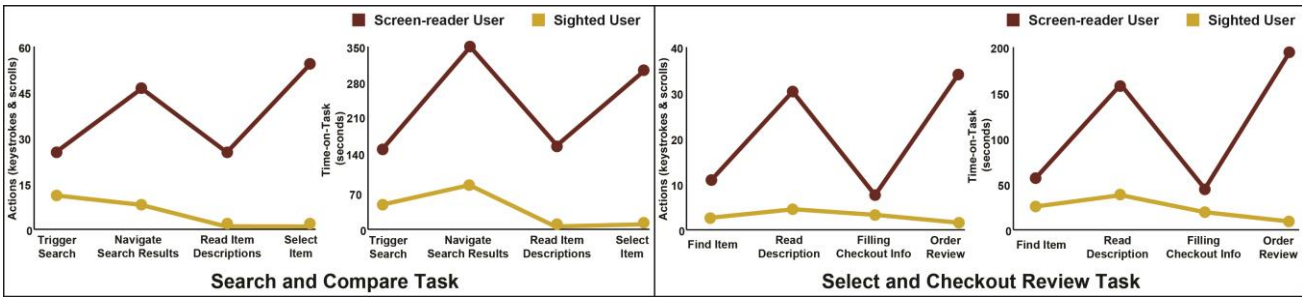


Figure 3: Navigation efficiency divide between sighted and BVI: screen-readers require users longer and more mechanical actions to complete most basic web tasks.

While navigating through the checkout page, screen-reader users who wanted to find out what was just added to the cart tried different advanced features like section skipping, JAWS find, and navigation by heading before resorting to the serial navigation through every single page element. These navigation problems emerging from the participants' experience suggest that crucial details of the page content are very difficult to access even in a simple page structure.

4.2 Rigid Access to Different Page Types

While commenting on the difficulties encountered, participants expressed concern about the fact that, although different pages exhibit a different structure, the way screen readers operate is too rigid and does not account for the flexibility needed in certain navigation contexts. One participant said:

> *[P4] "I have to navigate through different pages using the same approach. There are some pages like this [checkout page] that can be cut short [in presenting the content]. Unfortunately, we don't have any sophisticated options."*

Whereas a quick scroll and glance helps sighted users identify the context and important page sections to easily advance through them, screen readers are agnostic of the page type the user is browsing (e.g., a list page, a product detail page or an order review page), thus hampering an efficient navigation. In fact, participants voiced concerns about the consistent linear strategy of the screen readers and expressed the need for screen reader to somehow operate differently based on different page types.

4.3 Fear of Missing Crucial Information

> *[P2] "I fear that I might lose out important information on this page."*

Comments of this nature emerged frequently during the thinking aloud observation with participants. Especially when dealing with product orders and transactions, participants spent much longer time in navigating back and forth among sections and paid extra attention to the page details for the fear of either missing important information or engaging a link that might end in an unwanted action.

As shown in Figure 3, there is a large divide between the sighted and screen-reader users in terms of the actions made and time on task, especially on the item description and checkout pages. These are the two pages on which users must decide to pick a product of interest and double-check its details before placing their order, respectively. Whereas sighted users found no problem in quickly glancing through the page and double-check crucial information such as the product specification, screen-reader users spent more time going back and forth through the page elements to double-check the details.

Without the opportunity to aurally master "at a glance" the main purpose and content of the page, users are left searching and re-searching for page elements to make sure that important information are not missed.

4.4 Little Tolerance of Navigation Mistakes

Across all sessions, we observed that the screen readers offer little support to recover from a mistake. If a user mistakenly skips an important section, the whole process of scanning through the sections must be carefully repeated. For example, a participant looking for customer reviews used heading navigation to quickly look through the long list of sections on the item description page. In his first attempt, he passed the reviews section but did not realize it, and moved on to the end of the page. Once he realized that he was too far along and that he might have missed the reviews, he resorted to keying up and down each single page section until he re-found the reviews. Even for this basic task, recovering from this mistake slowed him down and required much longer and more effort than necessary.

This pattern is not just confined to user-made errors, but it can also result from labeling mismatch, excessive page content, and even the faulty interpretation of a screen reader. For example, when a participant wanted to find the product description on the checkout page, he unsuccessfully used *JAWS find* to match the information, because there is no such heading on that page. Then he used heading navigation and eventually landed on the order summary. It took awhile for him to reorient and start reading through each line on that page.

Whereas sighted users continuous monitored the overall page composition to locate the sought-after information, the screen reader impeded BVI users to exit the serial stream of links and page segments to quickly re-locate missed information from a higher-level perspective.

4.5 Blurred Sections and Overloaded Pages

> *[P4] "All the content around the main content is confusing. When JAWS reads the recommended products before reading out the full details of the product, I feel I am lost. I am not even sure about the title of the product I am in."*

Shopping websites features more and more often several *non-primary* sections, like *customers who bought this item also bought, frequently bought together, special offers and promotion*. In our study, we noticed that this ancillary content, although important for the site owners to promote up-selling and cross-selling, often confuses screen-reader users, because it makes it difficult to identify and access the most basic, fundamental functions of the page (*Add to Cart*). With no efficient way to work around such overload of information, users often got confused and made unnecessary steps.

We also observed that non-optimal organization of heading levels confused screen-reader users, causing a perception of *section blurring*: screen-reader users often found it difficult to *distinguish* between the available page sections and their current position on the page, thus forcing them to arrow down to each line of the content to serially scan the page.

Overall, overloaded pages and confusing section boundaries often led users to the most comfortable but inefficient mode of reading each page element, a strategy that causes unnecessary mechanical and cognitive effort for long and information-rich pages.

4.6 'Skipping Content' Not Always Optimal

Screen reader users can rely on a "*Skip to content"* link available on many websites to skip main navigation headings and menus and go directly to the main content section of a page. However, it can also cause serious navigation problems, such as the following critical incident observed in our user study:

A search string *'Dell Laptop'* on Amazon yielded *2346 results*. Amazon offers a *powerful faceted search* on the left-hand navigation panel. On the one hand, by using this function, our *sighted users quickly filtered the results down to six laptops* and accessed the product detail page in *three clicks*. However, *BVI users* did not realize the existence of the faceted search section because they "skipped" top the main content of the page. There is currently no option for the screen-reader user to facilitate the discovery of this very important function for basic search tasks.

4.7 Overloading Short-Term Memory

When asked to double-check the product details on the checkout page, a participant was able to recall them from the memory. He said:

> *[P5] "It is easier for me to remember such details than trying to find them with a screen reader."*

We observed that participants often employed external aides to save the details they need for easy re-finding at a later point in time. For example, when asked to compare different models of laptops and select one they liked, users browsed through the search results and copied the links of the products they liked in a separate file, along with a summary of product descriptions. Later in the tasks, when it was time to finalize their purchasing decision, they went back to the notepad to compare their findings and select one from there.

This behavior suggests that providing more immediate access to important details would reduce the mechanical and cognitive load now necessary to retain content details in short-term memory.

4.8 Preferences for Page-level Voice Access

At the end of the study, we asked participants to gauge their willingness to accept voice input as a modality facilitating a direct access structure. BVI users who were already accustomed to using Siri [25] or Talkback [26] on their smartphones reported feeling comfortable using voice provided the technology works as intended.

> *[P7] "I feel very comfortable using voice navigation. I always use Siri on my iPhone."*

Participants, however, expressed concern about using voice commands in public places especially when dealing with confidential details (purchasing details) or out of respect for privacy. A participant expressed his inclination toward direct access structures with no preference for the input modality, *"I am less concerned about the mode of input. I look forward to options to navigate through different sections quickly."* Participants proposed alternative access structures like predefined key combinations to help during technological failures and for use in noisy environments:

> *[P6] "I would love to have some key combinations [shortcuts] also for voice commands. This will give me a chance to use them when I don't want to use voice commands or just in case when the technology doesn't work."*

Participants expressed interest in working with a system that can be controlled with a very small set of short voice commands (easy to remember) as an additional option to support more direct navigation across page sections.

In sum, the results of our formative study provide ample evidence of the need for having more direct access to highly-relevant page sections to accelerate navigation for screen-reader users.

5. CROWD RANKING PAGE SECTIONS

How can a group of "important" page sections be determined, given the variety and richness of modern websites? It is clear that section relevance is tied to the main user's task supported by the purpose of a page. For example, search-filtering options may still be available on the product detail page, but they are not directly supporting the main purpose of the page (browsing product-related information). Because of the complexity of the page structure of large websites, it is essential to collect reliable data from users to identify the most important sections of a preliminary set of recurring page types based on common navigation tasks.

Prioritizing relevant page sections for common tasks can inform strategies to make highly-relevant sections quickly available on-demand. To this end, we designed and executed a crowdsourcing study to elicit an appropriate task-based ranking of the sections of *four page types* on Amazon.com: *home, search results, product details, and checkout pages.* For this study, we developed a custom, crowd-based section ranking system that allows crowd workers to quickly rank order different page section via drag and drop in the context of assigned navigation tasks.

5.1 Participants

We recruited 100 participants through Amazon Mechanical Turk (AMT) [27]. To maximize the internal validity of the results, we restricted the pool of crowd workers to US residents with at least 95% acceptance rate for the Human Intelligence Tasks (HIT) completed. To ensure quality of the rankings, we tracked the time users spent on each page and the number of scrolls for each HIT. HITs with valid comments and at least 20 minutes elapsed time were accepted. In some cases when we could not decide based on the above criteria, we used the scrolling count as an extra measure to accept a HIT. We only accepted data from 40 of 95 HITs that complied with our quality guidelines. For an average 30 minutes of participation, each participant received a $3 payment through AMT platform.

5.2 Procedure and Apparatus

The HIT asked participants to rank 60 sections (including content and navigation elements) spread over nine different page instances based on their importance in helping with the purchasing decision and the checkout process. The assigned shopping tasks – properly split in sub-tasks and contextually presented across the page instances – were the same as the search, compare and checkout tasks used in the formative study. For each ranking, participants were asked also to provide page-level comment to justify their preferences. On average, it took 30 seconds to rank each section (~30 min for the overall HIT).

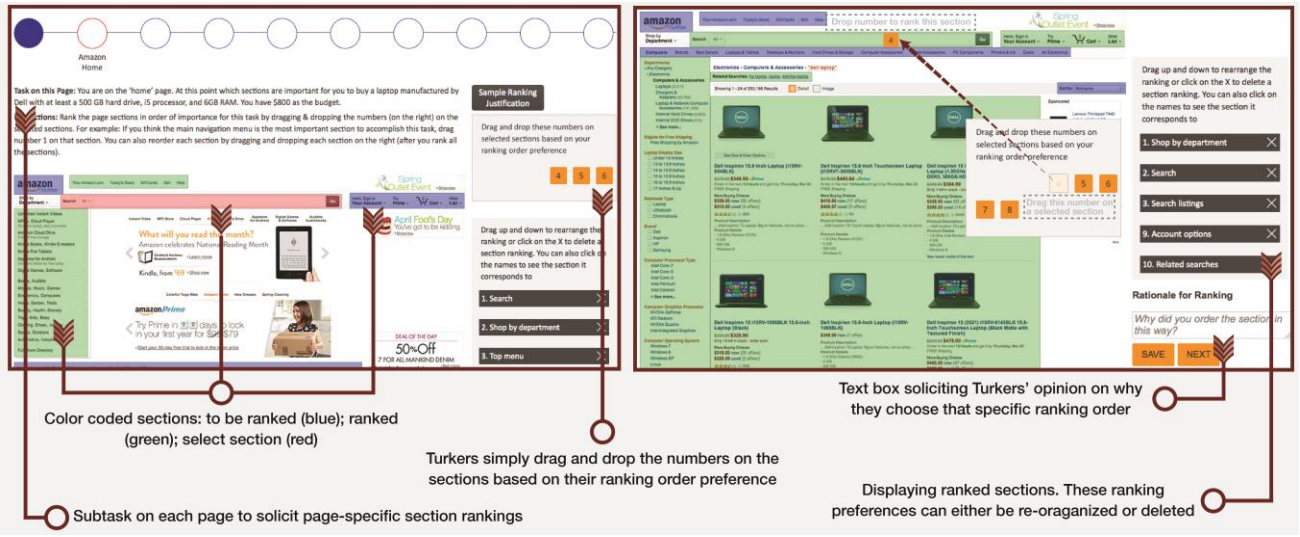


Figure 4. Our crowd-ranking system elicited from users the prioritized importance of different page sections for a common shopping task across nine instances of five different page types.

5.2.1 Custom Crowd-Ranking System

We developed a web-based interactive system [27] to facilitate this ranking process. The system (Figure 4) provides to participants a flow of screenshots simulating the navigation steps of Amazon.com. In the context of an assigned shopping task, the system enabled participants to drag and drop the rank numbers onto a matching section on the web page based on their preference. Participants could change or re-arrange the ranking as many times as they wish before submitting. At the end of the study, we gave them a reference code for validating completion and approving payment.

5.3 Analysis and Results

We calculated the mode of each ranked section in the nine different page types. Because of the rank-order nature of the collected data, we considered the mode as the most measure to determine the level of importance of a given section.

Section	Mean	Median	Mode	Our Rank
Search Listings	1.52	1	1	1
Sorting Options	3.10	2	3	2
Comment: I can drill down all the details to find my specific laptop, and then find everything I need on the first result page; all other options are not as important as this one.				
Shop by Department	3.82	3	3	3

Figure 5. When two sections in a page scored the same mode, we used mean, median, and user's comments to settle discrepancies

Figure 6 shows the *top-five ranked sections* for each of the four page types studied. We compared this ranking to the actual position of the page section in the linear reading strategy currently available via JAWS, based on input and manual calculation from one expert screen-reader user. The difference between the two rankings indicates that sections that users deemed most crucial for their task (and should, therefore, be immediately available) are inefficient to access (require several keystrokes) for expert screen-reader users. This gap can be even wider for non-expert users.

For example, in the *checkout* page, the shipping speed was considered among the five most important sections by our crowd participants, and it is also very easy to spot for sighted users at a glance. However, to get to the shipping speed, an expert screen-reader user must execute 22 keystrokes.

Page Type	Crowd-based ranking (most important sections first)	No. of screen-reader actions (Keystrokes)
 Home	Search	1
	Shop by Department	5
	Top Menu	9
	Account Options	4
	Recommended Sections	4
 Search Results	Search Listings	2
	Sorting Options	11
	Shop by Department	5
	Search	1
	Item Menu	18
 Item Description	Item Summary	3
	Product Information	8
	Customer Reviews	11
	Customer Questions	12
	Cart Options	6
 Added to Cart	Product Summary	2
	Checkout Options	1
	Customers Also Bought	4
	Financing	10
	Recommended Sections	6
 Checkout	Shipping Address	7
	Order Summary	1
	Product Summary	7
	Payment Method	12
	Shipping Speed	22

Figure 6. The serial navigation paradigm of screen readers buries highly-important sections necessary to complete common web tasks.

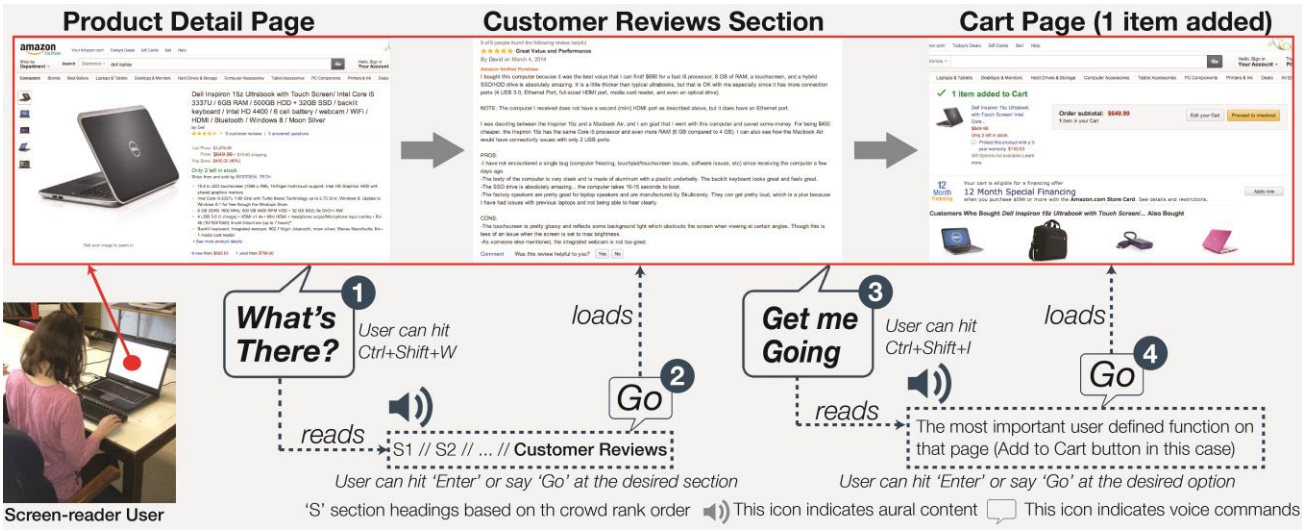


Figure 6. Through simple commands, DASX foregrounds on-demand manageable groups of key sections for different page types. By freeing users from serially listening to page content, DASX has the potential to reduce mechanical effort and access time to target relevant content.

6. AURAL GLANCING WITH DASX

Based on the results of our studies, we envision a *Dialogic Augmentation of the Screen-reader Experience (DASX)*, a novel navigation approach for instant, on-demand access to relevant sections. With DASX, screen-reader users can rely on a simple, high-level vocabulary of commands (via *key combinations* or *voice commands*) to "aurally glance" at a page by quickly invoking and browsing the most important sections available. This will enables user to bypass the heavy keyboard usage and segments scanning typical of screen-reader interaction. DASX is based on the following set of commands:

- **What's there** ***(Ctrl+Shift+W):*** *Reads the top-priority, five main sections of the page* (see ranking in Figure 6).
- **More on this page** ***(Ctrl+Shift+W):*** *Reads the other sections of the page.*
- **Get me going** ***(Ctrl+Shift+I):*** *Reads the one, most important function on the page (e.g., "Add to Cart" for shopping site, or it goes to the product page of the first item in a list)*
- **Go** ***(Enter):*** *Loads the desired option*

By augmenting, rather than replacing, screen-reader access, DASX foregrounds the five most relevant sections of a given page type (Figure 6) and makes them available to screen-reader users on-demand when they invoke a simple "What's There" command. The user may choose one of the five sections with "Enter" or "Go," or proceed to the next five by invoking "More on this page." Once the user decides to move on with an option s/he can say *Go* or hit *Enter* to load that section directly. The foreseen advantage for the user is the ability to efficiently get an overall sense of the content available (and purpose of the page) without being forced to arrow down through the detailed page segment. It is worth noticing that because DASX groups and makes available key page sections in collections of five, users do not risk missing other peripheral or ancillary content, accessible in subsequent section groups via "More on this Page."

To mitigate information overload, with a simple command "Get Me Going," DASX provides screen-reader users the ability to *activate at any time* the one, most important operation in a given page. For example, in the product detail page, "Get Me Going" would activate the "Add to Cart" button; in the search results page, "Get Me Going" would activate the product detail page of the first product in the list.

To quantify and assess the potential advantage of DASX in bridging the gap between sighted and screen-reader users, we conducted a preliminary, *analytical evaluation* of an expert user's task performance on DASX using KLM-GOMS [8].

In terms of evaluation parameters, keyboard and voice were selected as input modalities. Display and audio were selected as output modalities. We added *wait times* (5 s) for each step to account for the lag time a real system might encounter. All other operators were kept standard. We set the aural output at 350 words per minute, as this was the most common consumption rate of screen-reader users we observed in our formative study. To ensure comparable results we used the same tasks used during our formative study on *item description* and *checkout* pages. The sequence of steps followed is listed below.

Step 1: Launch DASX, which is manifested as a browser plug-in

Step 2: Access important sections with *What's there?*

Step 3: Select a section of interest with *Go*

Step 4: Repeat steps 2 and 3 until the target information is reached

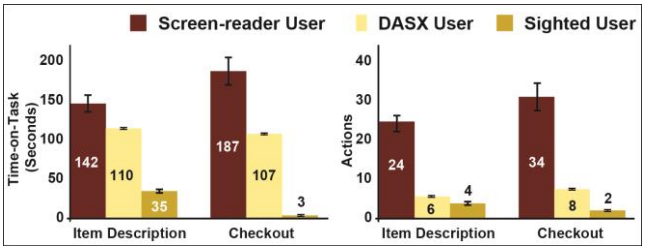


Figure 7. Expert performance suggests that DASX shortens the gap in page navigation efficiency between sighted and screen-reader users.

DASX drastically reduced the time and steps required on both *item description* and *checkout pages* as compared with the serial access of existing screen readers. With DASX, the time screen-reader users spent reviewing the order on the checkout page fell 43%. It fell only 23% on the item description page, because even with DASX screen-reader users need to listen to and attend to

product description and customer reviews. DASX not only decreased the number of actions required by screen-reader users by 75% on the item description page and by 69% on the checkout page but also brought their performance much closer to that of sighted users (Figure 7).

In absolute task performance measures for the "item description" and "checkout" pages, expert users with DASX require 6 and 8 keystrokes, which are very close to the 4 and 2 required by sighted users not using screen readers. This performance is in stark contrast with the 24 and 34 keystrokes currently necessary by screen-reader users to access the most important page sections. Such preliminary results are encouraging and confirm the potential for supporting a form of "aural glancing" to make sense of the page sections available.

7. CONCLUSIONS AND FUTURE WORK

The serial nature of screen-reader navigation can be complemented by an accessible form of "aurally glancing" that enables BVI to quickly identify and access key sections on demand at any time. In this paper, we presented a formative study with BVI and sighted users that uncovered and characterized basic page-level navigation problems while completing basic shopping tasks. Based on these findings, we presented a crowd-ranking system and study that elicited users' task-based preferences for the most important sections of recurrent page types. With these results, we presented and preliminarily validated DASX, a navigation approach that empowers screen-reader users to access directly the most important sections of a page via a set of high-level, always available commands. By supporting a form of aural glancing, the main contribution of our approach is to suggest strategies to drastically reduce the mechanical effort of screen-reader users in navigating complex site pages, thus shrinking the gap between sighted and BVI users in performing common tasks.

Our future work pursues two main paths. First, we are conducting participatory design sessions with expert screen-reader users and BVI rehabilitation instructors at BOSMA enterprises [29] to further validate our crowd-ranked sections and expand the DASX vocabulary. Second, we are developing DASX as a suite of accessible browser plug-ins that receive ranked sections data from two sources: (a) an extended crowd-ranking system that can streamline the labeling process for any website; b) web designers who can directly flag (in HTML and CSS) highly-relevant sections of the page at design time.

8. ACKNOWLEDGMENTS

We thank BOSMA enterprises and all study participants for their time and contribution; J. George-Palilonis, A. Voida, and S. Voida for their feedback on the paper; C. Meyer and C. Johnson for their help in executing the studies. This research is based upon work supported by NSF grant IIS-1018054, an NSF REU supplement, and a Google Faculty Research Award.

9. REFERENCES

[1] Ahmed, F., Borodin, Y., Soviak, A., Islam, M., Ramakrishnan, I. V., & Hedgpeth, T. (2012). Accessible skimming: faster screen reading of web pages. In *Proc. of the 25th annual ACM symposium on User interface software and technology* (pp. 367-378). ACM.

[2] Asakawa, C., & Itoh, T. (1998). User interface of a home page reader. In *Proc. of the third international ACM conference on Assistive technologies* (pp. 149-156). ACM.

[3] Borodin, Y., Bigham, J. P., Stent, A., & Ramakrishnan, I. V. (2008). Towards one world web with HearSay3. In *Proc. of International Cross-Disciplinary Conference on Web Accessibility* (p. 128–129).

[4] Borodin, Y., Bigham, J. P., Raman, R., & Ramakrishnan, I. V. (2008). What's new?: making web page updates accessible. In *Proc. of the 10th international ACM SIGACCESS conference on Computers and accessibility* (pp. 145-152). ACM.

[5] Bjurstrom, H., Granberg, C., Hogberg, J., Johannsen, B., & Mcglashan, S. (2003). *U.S. Patent No. 6,594,348*. Washington, DC: U.S. Patent and Trademark Office.

[6] Cai, D., Yu, S., Wen, J. R., & Ma, W. Y. (2003a). Extracting content structure for web pages based on visual representation. In *Web Technologies and Applications*, 406-417.

[7] Cai, D., Yu, S., Wen, J. R., & Ma, W. Y. (2003b). *VIPS: a vision-based page segmentation algorithm*, 28. Microsoft tech report, MSR-TR-2003-79.

[8] Card, S., Moran, T., & Newell, A. (1983). The psychology of human-computer interaction. Hillsdale, NJ: Lawrence Erlbaum Associates.

[9] De Witt, J. C., & Hakkinen, M. T. (1998). Surfing the Web with pwWebSpeak. In *Proc. of the Technology and Persons with Disabilities Conference*.

[10] Gadde, P., & Bolchini, D. (2013). WebNexter: dynamic guided tours for screen readers. In *Proc. of the adjunct publication of the 26th annual ACM symposium on User interface software and technology* (pp. 85-86). ACM.

[11] Hanson, V. L., Richards, J. T., & Lee, C. C. (2007). Web access for older adults: voice browsing?. In *Universal Access in Human Computer Interaction. Coping with Diversity* (pp. 904-913). Springer.

[12] Harper, S., & Bechhofer, S. (2007). SADIe: Structural semantics for accessibility and device independence. ACM Transactions on Computer-Human Interaction (TOCHI), 14(2), 10.

[13] Islam, M. A., Ahmed, F., Borodin, Y., & Ramakrishnan, I. V. (2012). Thematic organization of web content for distraction-free text-to-speech narration. In *Proc. of the 14th international ACM SIGACCESS conference on Computers and accessibility* (pp. 17-24).

[14] Islam, M. A., Ahmed, F., Borodin, Y., Mahmud, J., & Ramakrishnan, I. V., (2010). Improving Accessibility of Transaction-Centric Web Objects. *SDM'10*.

[15] Kohlschütter, C., & Nejdl, W. (2008). A densitometric approach to web page segmentation. *In Proc. of the 17th ACM conference on Information and knowledge management*, 1173-1182.

[16] Poon, J., & Nunn, C. (2001). Browsing the Web from a Speech-based Interface. In *Proc. of INTERACT* (pp. 302-309).

[17] Raman, T. V. (2008). Specialized browsers. In *Web Accessibility* (pp. 195-213). Springer London.

[18] Rohani Ghahari, R., Ferati, M., Yang, T., & Bolchini, D. (2012, October). Back navigation shortcuts for screen reader users. In *Proc. of the 14th international ACM SIGACCESS conference on Computers and accessibility* (pp. 1-8). ACM.

[19] The Voice Browser Working Group. (2014). Retrieved from http://www.w3.org/Voice/

[20] Yang, T., Gadde, P., Morse, R., & Bolchini, D. (2013). Bypassing lists: accelerating screen-reader fact-finding with guided tours. In *Proc. of the 15th International ACM SIGACCESS Conference on Computers and Accessibility* (p. 7). ACM.

[21] Yesilada, Y., Harper, S., Goble, C., & Stevens, R. (2004). Screen readers cannot see. In Web Engineering (pp. 445-458). Springer.

[22] Web Content Accessibility Guidelines (WCAG) 2.0. (2008). Retrieved from http://www.w3.org/TR/WCAG20

[23] Zhu, S., Sato, D., Takagi, H., & Asakawa, C. (2010). Sasayaki: an augmented voice-based web browsing experience. In *Proc. of the 12th international ACM SIGACCESS conference on Computers and accessibility* (pp. 279-280). ACM

[24] http://www.opera.com/help/tutorials/voice/using

[25] http://www.apple.com/ios/siri/

[26] http://developer.android.com/design/patterns/accessibility.html

[27] https://www.mturk.com/mturk

[28] Crowd-ranking site: http://tinyurl.com/crowdranking
Source Code: http://tinyurl.com/codeassets14

[29] http://www.bosma.org

[30] http://www.noldus.com/ulog/feature